

USB to UART module

USB to UART converter module convert USB data getting from connector and then convert to UART using FT232RL IC.

Hardware components used

FT232RL (28 pin package) from FTDI

3 pin connector for USB bus power with selectable external supply so one can select the supply between 3.3V or 5V DC

Ferrite bead

2 smd led for TX and RX data transmission indication with 330ohm resistor for led protection

6 pin MCU connector which is for uart connection

USB-B (USB TYPE-B) USB 2.0 Receptacle Connector 4 Position Through Hole

resistors

Capacitors

PINOUT OF FT232RL

USBDP= USB data signal plus

USBDM= USB data signal minus

gnd= device ground supply pins

VCC=3.3V to 5.25V input supply range

3v3out= this IC has internal voltage regulator which gives output of 3.3v

Connect this pin to ground by using 100nF

vccio= pin to decide which voltage value the i/o pins should be use

Either 3.3v or 5v

This is used when there is no MCU fixed used

3.3V logic= when vccio pin is connected to 3v3out pin

5 v logic= when vccio pin is connected to vcc

AGND= analog ground supply for internal clock multiplier

DTR= data terminal ready

RTS= handshake signal = it means ready to send

DSR= data set ready

CTS= clear to send

Test pin= puts ic into test mode

For normal connection test pin is connected with gnd

RTS(request to send)

Used to prevent the data loss when one device is faster than other
Used for handshaking before sending and receive the data with MCU

When ic wants to send data it creates RTS low which tells the MCU that I want to send data

If MCU is ready to receive the data CTS(clear to send)=low

USb power pin with= vcc of ic but with ferrite bead si that it can reduce noise

USB CONNECTOR PINOUT

vcc= the main input power supply

D+= for data +
Connect with MCU USB data + pin

D- = for data -
Connect with MCU USB data - pin

gnd= direct connect to ground

Shield= metal part body of the USB connector which will collect the electric discharge from the human touch or environment

Why shield pin should not connect direct to gnd

As the shield pin is the metal core body part which will collect the electric discharge from the human touch or environment so sudden discharge can cause voltage drop
there fore to reduce the votlage drop the resistor and capacitor parallel connection is used

Shield pin can connect directly to gnd but good method is R+C combination in parallel connection with shield then to gnd

capacitor= used to prevent high frequency EMI noise so EMI noise can goes to gnd

resistor= slowly removes the static charge from the shield

Ferrite bead

It converts high frequency noise into heat

Placed in series with vcc line

Used in switching regulator LDO, USB

Placed near with connector like USB type C is used then it should place near with connector

Decoupling capacitor at 3v3out pin

Reason of choosing FT232RL ic

The usb to uart conversion can be done also by CH340 ic or CP2102 ic

Advantages of FT232RL

- Stable drivers for macos, windows, linux so no need to install manually drivers while in other ic there is need of downloading manual driver for older system
- High tolerance for EMI noise and other signals
- Configurable via EEPROM
- Robust to use in harsh environment also
- Well defined Internal structure

Disadvantage of FT232RL

Very high price compare to other ic

UART connector

1X6 pin header 2.54mm pitch distance connector for UART